

YZV202E Optimization for Data Science (Homework 3)

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Question 1: Water Distribution Transportation Problem

Parameters

From student ID 150220322, the last four digits are 0, 3, 2, 2: - a = 4th from last = 0 → **5** (0 replaced by 5) - b = 3rd from last = 3 → **3** - c = 2nd from last = 2 → **2** - d = last digit = 2 → **2**

Unit costs (×1000 TL): Ankara→Çanakkale = 5, Ankara→Denizli = 3, Bursa→Çanakkale = 2, Bursa→Denizli = 2. Supply: Ankara = 90, Bursa = 80. Demand: Çanakkale = 55, Denizli = 65.

1.1 LP Formulation

Decision variables: x_{11} (Ankara→Çanakkale), x_{12} (Ankara→Denizli), x_{21} (Bursa→Çanakkale), x_{22} (Bursa→Denizli).

Minimize:

$$\min 5x_{11} + 3x_{12} + 2x_{21} + 2x_{22}$$

Subject to:

$$x_{11} + x_{12} \leq 90 \quad (\text{Ankara supply})$$

$$x_{21} + x_{22} \leq 80 \quad (\text{Bursa supply})$$

$$x_{11} + x_{21} = 55 \quad (\text{Çanakkale demand})$$

$$x_{12} + x_{22} = 65 \quad (\text{Denizli demand})$$

$$x_{11}, x_{12}, x_{21}, x_{22} \geq 0$$

1.2 Standard Form

Slack variables $s_1, s_2 \geq 0$ are introduced for the supply inequalities, converting them to equalities:

$$\min 5x_{11} + 3x_{12} + 2x_{21} + 2x_{22} + 0s_1 + 0s_2$$

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 90 \\ 80 \\ 55 \\ 65 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 5 \\ 3 \\ 2 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

1.3 Analytical Solution

Using the Minimum Cost Method, routes are allocated in ascending cost order:

Step	Route	Flow	Remaining
1	Bursa→Çanakkale (cost 2)	$x = 55$	Çanakkale satisfied, Bursa = 25
2	Bursa→Denizli (cost 2)	$x = 25$	Bursa exhausted, Denizli = 40
3	Ankara→Denizli (cost 3)	$x = 40$	Denizli satisfied, Ankara = 50
4	Ankara→Çanakkale (cost 5)	$x = 0$	Already satisfied

Optimal solution: $x = 0$, $x = 40$, $x = 55$, $x = 25$

Total cost: $5(0) + 3(40) + 2(55) + 2(25) = 0 + 120 + 110 + 50 = \mathbf{280 \times 1000 \text{ TL} = 280,000 \text{ TL}}$

This solution is optimal. Shifting any unit away from the Bursa routes (cost 2) toward Ankara→Çanakkale (cost 5) adds 3 units of cost per unit moved. There is no cheaper alternative.

1.4 CVXPY Validation

CVXPY confirms: optimal cost = 280,000 TL, with the same allocation ($x = 0$, $x = 40$, $x = 55$, $x = 25$).

1.5 Effect of Parameter a

As a varies from 1 to 10, two regimes emerge:

- **$a = 2$:** Ankara→Çanakkale is as cheap or cheaper than Bursa→Çanakkale (cost 2), so some flow shifts to x . The total cost drops as a decreases below 2.
- **$a = 2$:** Both cities cost the same to reach Çanakkale. The solver can split flows freely without affecting the total cost.

- **a > 2:** Ankara→Çanakkale is strictly more expensive. The optimal solution drops x_3 to zero - Bursa handles all of Çanakkale's demand ($x_2=55$), and Ankara only covers Denizli ($x_1=40$). Since a no longer appears in the active solution, the total cost stays flat at 280 regardless of how large a gets.

So the cost function is piecewise linear in a and completely flat for $a \geq 2$. Once a route is priced out of the solution, changes to its cost have no effect at all. This is a structural property of transportation problems.

Question 2: LP Duality

Primal:

$$\max x_1 + x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 3, \quad x_1 \geq x_2, \quad x_1, x_2 \geq 0$$

2.1 Dual Problem

Rewriting $x_1 \geq x_2$ as $-x_1 + x_2 \leq 0$, we assign dual variable $y_1 \geq 0$ to the first constraint and $y_2 \geq 0$ to the second:

Dual:

$$\begin{aligned} \min \quad & 3y_1 \\ & y_1 - y_2 \geq 1 \quad (\text{column for } x_1) \\ & 2y_1 + y_2 \geq 1 \quad (\text{column for } x_2) \\ & y_1, y_2 \geq 0 \end{aligned}$$

Derivation steps: - Objective: $\min \mathbf{b}^T \mathbf{y} = \min 3y_1 + 0 \cdot y_2 = \min 3y_1$ - For x_1 : A column = $[1, -1]^T$, $c_1 = 1 \rightarrow y_1 - y_2 \geq 1$ - For x_2 : A column = $[2, 1]^T$, $c_2 = 1 \rightarrow 2y_1 + y_2 \geq 1$

2.2 CVXPY Solution

Primal vertices: - $(0, 0)$: $f = 0$ - $(3, 0)$: $f = 3$, satisfies $x_1 \geq x_2$ - $(1, 1)$: $f = 2$, satisfies $x_1 = x_2$

→ **Primal optimal:** $x_1^* = 3, x_2^* = 0, \text{ value} = 3$

Dual check at $y_1=1, y_2=0$: $3(1)=3, y_1 - y_2 = 1 \geq 1, 2(1)+0=2 \geq 1$

→ **Dual optimal:** $y_1^* = 1, y_2^* = 0, \text{ value} = 3$

Strong duality holds: primal = dual = 3.

2.3 Effect of Removing $x = x$

Without the constraint, the feasible region gains the vertex $(0, 1.5)$, and the new optimal is still $(3, 0)$ with value **3** - identical to before.

The constraint was **inactive** at the original optimum ($x = 3 > 0 = x$). By complementary slackness, an inactive constraint carries a dual variable of zero, which means it puts no pressure on the objective. Removing it changes nothing. This is the practical takeaway: only binding constraints actually shape the solution.

Question 3: Lagrange Multipliers

$$\min_{x,y} xy \quad \text{s.t.} \quad x^2 + y^2 = 2$$

(a = last digit of student ID = 2)

3.1 Contour Plot

The contour lines of $f(x,y) = xy$ are hyperbolas symmetric about both diagonals. The constraint is a circle of radius $\sqrt{2}$. The minimum sits where a contour is tangent to the circle from below (most negative xy); the maximum is tangent from above.

At the minimum points $(1,-1)$ and $(-1,1)$, the gradient of the objective $f = [y, x]$ and the gradient of the constraint $g = [2x, 2y]$ point in opposite directions exactly the Lagrange condition. See Figure q3_contour.png.

3.2 Analytical Solution

Lagrangian: $L(x,y, \lambda) = xy + \lambda(x^2 + y^2 - 2)$

Stationarity conditions: - $L/\lambda = x + 2\lambda x = 0 \rightarrow y = -2\lambda x$ (1) - $L/y = x + 2\lambda y = 0 \rightarrow x = -2\lambda y$ (2) - $L/\lambda = x^2 + y^2 - 2 = 0$ (3)

Substituting (1) into (2): $x = -2(-2\lambda x) = 4\lambda^2 x$. Since $x \neq 0$: $4\lambda^2 = 1 \rightarrow \lambda = \pm 1/2$.

$\lambda = -1/2$: $y = x \rightarrow 2x^2 = 2 \rightarrow x = \pm 1 \rightarrow$ Points $(1,1), (-1,-1)$, $f = +1$ (maximum)

$\lambda = +1/2$: $y = -x \rightarrow 2x^2 = 2 \rightarrow x = \pm 1 \rightarrow$ Points $(1,-1), (-1,1)$, $f = -1$ (minimum)

3.3 Effect of a

For the general problem $\min xy$ s.t. $x^2 + y^2 = a$: - Optimal points: $(\sqrt{a/2}, -\sqrt{a/2})$ and $(-\sqrt{a/2}, \sqrt{a/2})$ - Minimum value: **$f^* = -a/2$**

The circle just scales with a . The optimal direction ($y = -x$ diagonal) never changes. Only how far out on that diagonal the solution lands. So doubling a doubles the magnitude of the optimal value.

3.4 scipy Verification

`scipy.optimize.minimize` (SLSQP) confirms $f^* = -1.000000$ at $(1, -1)$ and $(-1, 1)$.

3.5 Maximization vs Minimization

Maximizing xy on the same circle gives $f^* = +1$ at $(1, 1)$ and $(-1, -1)$. (the $y=x$ diagonal instead of $y=-x$.)

Along the circle parametrized as $(\sqrt{a} \cdot \cos \theta, \sqrt{a} \cdot \sin \theta)$, the objective is $f(\theta) = (a/2) \cdot \cos(2\theta)$. It oscillates between $-a/2$ and $+a/2$. The minimum is at $\theta = 3/4$ (and $7/4$), the maximum at $\theta = \pi/4$ (and $5/4$). Between them, the function is strictly monotone on each arc, there are no other stationary points on the constraint.

3.6 SONC, SOSC, and Local Minimizer Check

The Hessian of the Lagrangian is:

$$\nabla_{xx}^2 \mathcal{L} = \begin{bmatrix} 2\lambda & 1 \\ 1 & 2\lambda \end{bmatrix}$$

At the minimum point $(1, -1)$ with $\lambda = +1/2$:

$$\nabla_{xx}^2 \mathcal{L} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

The tangent space to the constraint at $(1, -1)$ is spanned by $g = [2, -2]$, so the tangent direction is $d = t[1, 1]$ for $t \in \mathbb{R}$. Evaluating:

$$d^T \nabla_{xx}^2 \mathcal{L} d = t^2 \cdot [1, 1] \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 4t^2 > 0 \quad \forall t \neq 0$$

SOSC is satisfied (strictly positive definite on the tangent space), so $(1, -1)$ is a strict local minimizer. By symmetry, $(-1, 1)$ is too. And since the constraint set is compact and we've found all stationary points, both are global minimizers.

SONC is also satisfied SOSC is the stronger condition and implies it.

Question 4: KKT Conditions

$$\min_{x_1, x_2} x_1^2 + x_2^2 \quad \text{s.t.} \quad g_1 : x_1 \leq 7, \quad g_2 : -x_1 + \frac{x_2^2}{4} + 4 \leq 0$$

4.1 Contour Plot and Feasible Region

The objective contours are concentric circles centered at the origin. But $g_2 = 0$ forces $x_1 = x_2^2/4 + 4 \geq 4$, so the entire feasible region lies to the right of $x_1 = 4$. The origin is not feasible.

The smallest circle touching the feasible region hits the parabola $g_2 = 0$ at $(4, 0)$ the point on that curve closest to the origin. Setting $x_2 = 0$ minimizes the distance, giving $x_1 = 4$. See Figure q4_contour.png.

Constraint activity at the optimum $(4, 0)$: - $g_1 = 4 - 7 = -3 < 0 \rightarrow$ **inactive** - $g_2 = -4 + 0 + 4 = 0 \rightarrow$ **active** (binding)

4.2 KKT Conditions

KKT Lagrangian:

$$\mathcal{L} = x_1^2 + x_2^2 + \mu_1(x_1 - 7) + \mu_2\left(-x_1 + \frac{x_2^2}{4} + 4\right)$$

Stationarity: - K1: $2x_1 + \mu_1 - \mu_2 = 0$ - K2: $2x_2 + \mu_2 \cdot (x_2/2) = 0$

Primal feasibility: $x_1 \leq 7, -x_1 + x_2^2/4 + 4 \leq 0$

Dual feasibility: $\mu_1, \mu_2 \geq 0$

Complementary slackness: $(x_1 - 7)\mu_1 = 0, (-x_1 + x_2^2/4 + 4)\mu_2 = 0$

4.3 Analytical Solution

Regularity (LICQ): Where g_1 is active, $\nabla g_1 = [1, 0]$ and $\nabla g_2 = [-1, x_2/2]$. These are linearly independent for $x_2 \neq 0$. At $x_2 = 0$ (which is the case we find), g_2 is inactive so only one constraint is active and LICQ holds trivially.

Case 1: g_1 inactive ($\mu_1 = 0$), g_2 active

From K2: $x_2(2 + x_2/2) = 0$.

Sub-case 1a ($x_2 = 0$): g_2 active $\rightarrow -x_1 + 4 = 0 \rightarrow x_1 = 4$. From K1: $8 - \mu_2 = 0 \rightarrow \mu_2 = 8 > 0$. Check $g_1: 4 - 7 = -3 < 0 \rightarrow$ **Valid KKT point: $(4, 0)$, $f = 16$**

Sub-case 1b ($x_2 \neq 0$): $\mu_2 = -4 < 0$. Violates dual feasibility. Invalid.

Case 2: Both g_1 and g_2 active ($x_1 = 7$)

$g_2: -7 + x_2^2/4 + 4 = 0 \rightarrow x_2^2 = 12 \rightarrow x_2 = \pm 2\sqrt{3}$. From K1 with $x_1 = 7$: $14 - \mu_1 - \mu_2 = 0$. $\mu_2 = -4 < 0$. Invalid.

Case 3: g active ($x = 7$), g inactive ($\lambda = 0$)

From K2: $x = 0$. From K1: $14 + \lambda = 0 \rightarrow \lambda = -14 < 0$. Invalid.

The only valid KKT point is $(\mathbf{x}^*, \mathbf{x}^*) = (4, 0)$ with $\mathbf{f}^* = 16$. Only g is active; g plays no role in the solution.

4.4 CVXPY Verification

CVXPY (CLARABEL solver) confirms: $f^* = 16.000000$, $x = 4.000000$, $x = 0.000000$.